THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE · NCERT-TYPE FULL-YEAR PAPER SET 2

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. In a dihybrid cross between round yellow (RRYY) and wrinkled green (rryy) seeds, the F1 seeds are 1

(a) wrinkled yellow **(b)** all round yellow
(c) all wrinkled green **(d)** round green

2. The refractive index of a medium is the ratio of the speed of light 1

(a) in the medium to that in vacuum **(b)** in glass to that in water only
(c) in air to that in water only **(d)** in vacuum to that in the medium

3. Myopia is corrected by using 1

(a) a plane mirror **(b)** a cylindrical lens
(c) a convex lens **(d)** a concave lens

4. Ohm's law states that the current through a conductor is 1

(a) directly proportional to the potential difference, at constant temperature
(b) proportional to the square of V
(c) independent of the potential difference
(d) inversely proportional to the potential difference

5. The field pattern of a solenoid is similar to that of 1

(a) a plane mirror **(b)** a straight wire
(c) a horseshoe magnet only **(d)** a bar magnet

6. A food chain always begins with 1

(a) producers **(b)** carnivores **(c)** decomposers **(d)** herbivores

7. $\mathrm{Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu}$ is a displacement reaction because: 1

(a) copper gets oxidised
(b) zinc is less reactive than copper
(c) both metals are exchanged as anions
(d) zinc is more reactive and displaces copper from its salt

8. Milk of magnesia, an antacid, is: 1

(a) magnesium hydroxide **(b)** magnesium sulphate
(c) magnesium carbonate **(d)** magnesium chloride

9. Metals like sodium and potassium are extracted from their ores by: 1

(a) roasting **(b)** electrolysis of molten chlorides
(c) heating alone **(d)** reduction with carbon

10. The property of carbon to form long chains by bonding with itself is called: 1

(a) allotropy **(b)** isomerism **(c)** catenation **(d)** polymerisation

11. The functional unit of the kidney is: 1

(a) alveolus **(b)** neuron **(c)** nephron **(d)** villus

12. The hormone responsible for the fight or flight response is: 1

(a) thyroxine **(b)** adrenaline **(c)** growth hormone **(d)** insulin

13. Sperms are produced in the 1

(a) testes **(b)** ureter **(c)** prostate **(d)** urethra

14. In human beings, the sex of the child is determined by 1

(a) the egg alone
(b) environmental temperature
(c) the type of sperm that fertilises the egg
(d) the mother's diet

15. A convex lens is also called 1

(a) a cylindrical lens **(b)** a converging lens
(c) a diverging lens **(d)** a plane lens

16. Splitting of white light into its seven colours is called 1

(a) scattering **(b)** refraction **(c)** dispersion **(d)** reflection

17. For resistors in series, the equivalent resistance is 1

(a) 1/R1 + 1/R2 **(b)** the average of the resistances
(c) less than each **(d)** R1 + R2 + R3

18. Electromagnetic induction is the production of 1

(a) heat by current
(b) induced current in a coil due to a changing magnetic field
(c) force on a coil
(d) a magnetic field by a steady current

19. The ozone layer is present in the 1

(a) hydrosphere **(b)** lithosphere **(c)** stratosphere **(d)** troposphere

20. In the reaction $\mathrm{Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe}$, the type of reaction is: 1

(a) Decomposition **(b)** Double displacement
(c) Combination **(d)** Displacement

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. Why is variation useful for a species? 2

22. What is the difference between heredity and variation? 2

23. State the laws of reflection of light. 2

24. What is meant by the power of accommodation of the eye? 2

25. Define one ampere of current. 2

26. State the rule to find the direction of the magnetic field around a straight current-carrying wire. 2

SECTION — C (Short Answer Type) — 6 × 3 = 18

Answer in about 40–60 words each.

27. List the characteristics of the image formed by a plane mirror. 3

28. Describe the structure and function of the cornea, iris, pupil and retina. 3

29. A current of 0.5 A flows in a conductor for 10 minutes. Find the charge that flows and the number of electrons. 3

30. Draw (describe) the magnetic field lines around a bar magnet and state three properties. 3

31. Explain the difference between a food chain and a food web. 3

OR

What is a balanced chemical equation? Balance $\mathrm{Fe + H_2O \longrightarrow Fe_3O_4 + H_2}$. 3

32. What is the chlor-alkali process? Name its three products and write the equation. 3

OR

How is zinc obtained from its carbonate ore? Write equations. 3

SECTION — D (Long Answer Type) — 6 × 5 = 30

Answer in about 100–150 words each.

33. Draw the structure of the human eye (describe) and explain the function of its parts: cornea, iris, pupil, lens, retina, optic nerve. 5

34. State and explain Ohm's law. Describe an activity to verify it and how the V-I graph is used to find the resistance. 5

35. Describe an activity to show that a current-carrying wire produces a magnetic field around it. State the right-hand thumb rule. 5

36. What is an ecosystem? Describe its biotic and abiotic components with examples. 5

OR

Define oxidation and reduction. Identify the substance oxidised, reduced, the oxidising agent and the reducing agent in $\mathrm{ZnO + C \longrightarrow Zn + CO}$. 5

37. Describe the chlor-alkali process. Write the equation, name the electrodes where gases are formed and give one use of each product. 5

OR

What is corrosion? Describe the conditions for rusting. State four ways to prevent it. 5

38. What is esterification? Write the reaction, the conditions, the smell and uses of esters. Write the saponification reaction. 5

OR

Describe excretion in human beings: parts of the system and the working of the nephron. 5

END OF QUESTION PAPER — Total: 80 Marks